



AKSHAYKUMAR GUNARI

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CAREER OBJECTIVE

Systems software engineer with experience in Linux kernel development, device drivers, and platform bring-up across server and networking hardware. Seeking opportunities to contribute to high-performance systems while expanding expertise in operating systems, kernel architecture, and hardware-software co-design.

EXPERIENCE

- Juniper Networks - HPE India Private Limited** Bangalore, India
Software Engineer - 3 July 2025 - Present 0 year(s) and 8 month(s)
 - Develop Linux kernel drivers for QFX-5K data center switches, contributing to hardware bring-up, platform debugging, and performance optimization to support reliable and high-throughput networking systems.
- Software Engineer - 2** March 2024 - June 2025 1 year(s) and 3 month(s)
 - Contributed to QFX-5K switch platform bring-up within the Linux Platform Development team.
 - Developed platform daemons on [Junos EVO OS](#) and debugged platform-level issues across system components.
- AMD India Private Limited** Bangalore, India
System Software Designer - 2 May 2023 - March 2024 0 year(s) and 10 month(s)
 - Developed Linux kernel drivers for DMA and Non-Transparent Bridge (NTB) subsystems on server platforms.
 - Built performance evaluation tools (integrating [dma-test](#) and [ntb_perf](#)) to measure DMA data transfer efficiency.
System Software Designer - 1 July 2021 - May 2023 1 year(s) and 10 month(s)
 - Enabled Human Presence Detection (HPD) and Ambient Light Sensors (ALS) on AMD platforms using an ARM Cortex-M4 co-processor.
 - Integrated sensors including TDK CH201 and Sunny M1, developing firmware and device drivers for sensor enablement for [AMD MP2](#).
- Centre of Excellence in Visual Intelligence** Hubli, India
Research Intern March 2021 - July 2021 0 year(s) and 4 month(s)
 - Worked on a DST-funded research project focused on clustering and categorization of crowdsourced 2D and 3D data
 - Co-authored 4 research papers published in national and international conferences.
- Indian Institute of Technology - Delhi** New Delhi, India
Project Trainee June 2019 - July 2019 0 year(s) and 2 month(s)
 - Worked on endoscopic tool tracking as part of a neuro-engineering research project. (RP03571G) [Internship Certificate](#)

EDUCATION

- Birla Institute of Technology** Pilani, India
Masters of Technology in Artificial Intelligence and Machine Learning May 2024 - Pursuing
 - GPA: Pursuing
- KLE Technological University** Hubli, India
Bachelor of Engineering in Computer Science August 2017 - July 2021
 - GPA: 9.02/10.00

PROJECTS

- NTB and DMA Driver Development: [Associated with AMD India Pvt Limited]** January 2023 - March 2024
Domains: [Linux Kernel, DMA, NTB]
 - Developed linux kernel drivers for DMA and Non-Transparent Bridge (NTB) on server platforms.
 - Built kernel modules for performance evaluation of NTB data transfer over DMA across multi-node systems.
- Designing and Enablement of Zephyr RTOS: [Associated with AMD India Pvt Limited]** March 2022 - December 2022
Domains: [RTOS, Zephyr, ARM Processors, Device Tree]
 - Ported and enabled Zephyr RTOS on ARM Cortex-M4, including device tree and memory configuration.
- HPD Sensors Enablement: [Associated with AMD India Pvt Limited]** July 2021 - December 2022
Domains: [Firmware, Sensors Enablement, I2C Protocol]
 - Enabled HPD and sensor stack on ARM Cortex-M4 platforms.
 - Developed firmware for sensor integration and data communication to OS services.
- Progressive Clustering: [Associated with Centre of Excellence in Visual Intelligence (CEVI)]** January 2021 - July 2021
Domain: [Deep Learning, Computer Vision, Incremental Learning, Unsupervised Learning]
 - Developed an unsupervised learning framework for clustering dynamically growing datasets with evolving class distributions.
- ABD-Net: [Associated with Centre of Excellence in Visual Intelligence (CEVI)]** March 2021 - June 2021
Domains: [Deep Learning, Computer Vision, 3D Vision, Point Clouds, Unsupervised Learning]
 - Developed Attention Based Decomposition - Network (ABD-Net), a deep learning model for 3D point cloud decomposition into basic geometric shapes.

- [C.1] Akshaykumar Gunari, et al. (2022). **Progressive Clustering: An Unsupervised Approach Towards Continual Knowledge Acquisition of Incremental Data**. In *ICPRAI 2022 - 3rd International Conference on Pattern Recognition and Artificial Intelligence*. (Conference held in Paris, June 01), pp. 355–367. Publisher. 01 June 2022, Paris. DOI: 10.1007/978-3-031-09282-4_30
- [C.2] Akshaykumar Gunari, et al. (2022). **ABD-Net: Attention Based Decomposition Network for 3D Point Cloud Decomposition**. In *The first Workshop on Structural and Compositional Learning on 3D Data - ICCV 2021*, IEEE Xplore. 16th October 2021, Held Virtually. DOI: 10.1109/ICCVW54120.2021.00232
- [C.3] Akshaykumar Gunari, et al. (2020). **Augmented Data as an Auxiliary Plug-In Toward Categorization of Crowdsourced Heritage Data**. In *Workshop on Digital Heritage (WDH) 2021*, pp. 53–62. Springer, Singapore. 22nd December 2021, IIT-Jodhpur, India. DOI: 10.1007/978-981-19-4136-8_4
- [C.4] Akshaykumar Gunari, et al. (2020). **Deep Visual Attention-Based Transfer Clustering**. In *Fourth International Conference on Computing and Network Communications (CoCoNet'20)*, pp 357–366. Springer, Singapore. 14th October 2020, Chennai, India. DOI: 10.1007/978-981-33-6987-0_29
- [P.1] Siddharth Katageri, Akshaykumar Gunari, Shashidhar V Kudari, et al. (2022). **System for Attention-Based Decomposition of 3D Point Clouds**. Registration Date: 18th March 2022.

SKILLS

- **Programming Languages:** C, C++, Python, Shell
- **Industry Knowledge:** Operating Systems, Linux Kernel Programming, Device Drivers, Yocto Project, RTOS, AI/ML, Computer Vision.

HONORS AND AWARDS

- **Spotlight Award** December 2021
AMD India Private Limited
◦ Delivered a critical feature (S2-Idle boot timestamp measurement) in the S2-Idle Cycle in Smart Trace Buffer (STB) Parser, required for Google Chromebook platforms.
- **Spotlight Award** December 2022
AMD India Private Limited
◦ Enabled TDK CH201 HPD sensor on AMD platforms, supporting integration of multi-sensor stack (accelerometer, gyroscope, magnetometer, ALS, HPD) on the co-processor.

Date: Thursday 26th March, 2026



(Akshaykumar Gunari)